Erratum to "Convex ordering for random vectors using predictable representation"

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In Theorem 4.1 of [1], the processes $(\hat{A}(t))_{t\in\mathbb{R}_+}$ and $(t,x) \mapsto \hat{B}_t(x)$ should be deterministic functions (and not $\mathcal{F}_t^M$-predictable processes) in order for the convex concentration inequality (4.2) to hold. The same applies to $(\hat{A}(t))_{t\in\mathbb{R}_+}$ and $(\hat{J}(t))_{t\in\mathbb{R}_+}$ in Corollaries 4.2 and 4.3.

Processes that are $\mathcal{F}_t^M$-predictable (instead of being deterministic functions) can be considered in (4.2) under additional conditions based on the Malliavin calculus, cf. [2] in the one-dimensional case with jumps.

References

[1] M. Arnaudon, J.-C. Breton, and N. Privault. Convex ordering for random vectors using predictable representation. *Potential Anal.*, 29(4):327–349, 2008.

[2] J.-C. Breton, B. Laquerrière, and N. Privault. Convex comparison inequalities for non-Markovian stochastic integrals. Preprint, to appear in Stochastics, 2012.

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